Power Sector at a Glance ALL INDIA

Updated on 30-11-2024
Source: OM SECTION

1.Total Installed Capacity (As on 30.11.2024) - Source : Central Electricity Authority (CEA)

INSTALLED GENERATION CAPACITY (SECTOR WISE) AS ON 30.11.2024

Sector	MW	% of Total
Central Sector	104452.93	24.6%
State Sector	108330.44	26.0%
Private Sector	243974.13	49.5%
Total	456757.49	

Installed GENERATION CAPACITY(FUELWISE) AS ON 30.06.2022		
CATAGORY	INSTALLED GENERATION CAPACITY(MW)	% of SHARE IN Total
Fossil Fuel		
Coal	2,04,080	50.7%
Lignite	6,620	1.6%
Gas	24,856	6.2%
Diesel	510	0.1%
Total Fossil Fuel	2,36,065	58.5%
Non-Fossil Fuel		
RES (Incl. Hydro)	1,59,949	39.7%
Hydro	46,850	11.6 %
Wind, Solar & Other RE	1,14,064	28.3 %
Wind	40,788	10.1 %
Solar	57,706	14.3 %
BM Power/Cogen	10,206	2.5 %
Waste to Energy	477	0.1 %
Small Hydro Power	4,888	1.2 %
Nuclear	6,780	1.7%
Total Non-Fossil Fuel	1,67,694	41.5%
Total Installed Capacity (Fossil Fuel & Non-Fossil Fuel)	4,03,760	100%

Total Generation and growth over previous year in the country during 2009-10 to 2022-23

Year	Total Generation (Including Renewable Sources) (BU)	% of growth
2009-10	808.498	7.56
2010-11	850.387	5.59
2011-12	928.113	9.14
2012-13	969.506	4.46
2013-14	1,020.200	5.23
2014-15	1,110.392	8.84
2015-16	1,173.603	5.69
2016-17	1,241.689	5.80
2017-18	1,308.146	5.35
2018-19	1,376.095	5.19
2019-20	1,389.102	0.95
2020-21	1,381.827	-2.49
2021-22	1.491.859	7.96
2022-23*	430.972	16.79

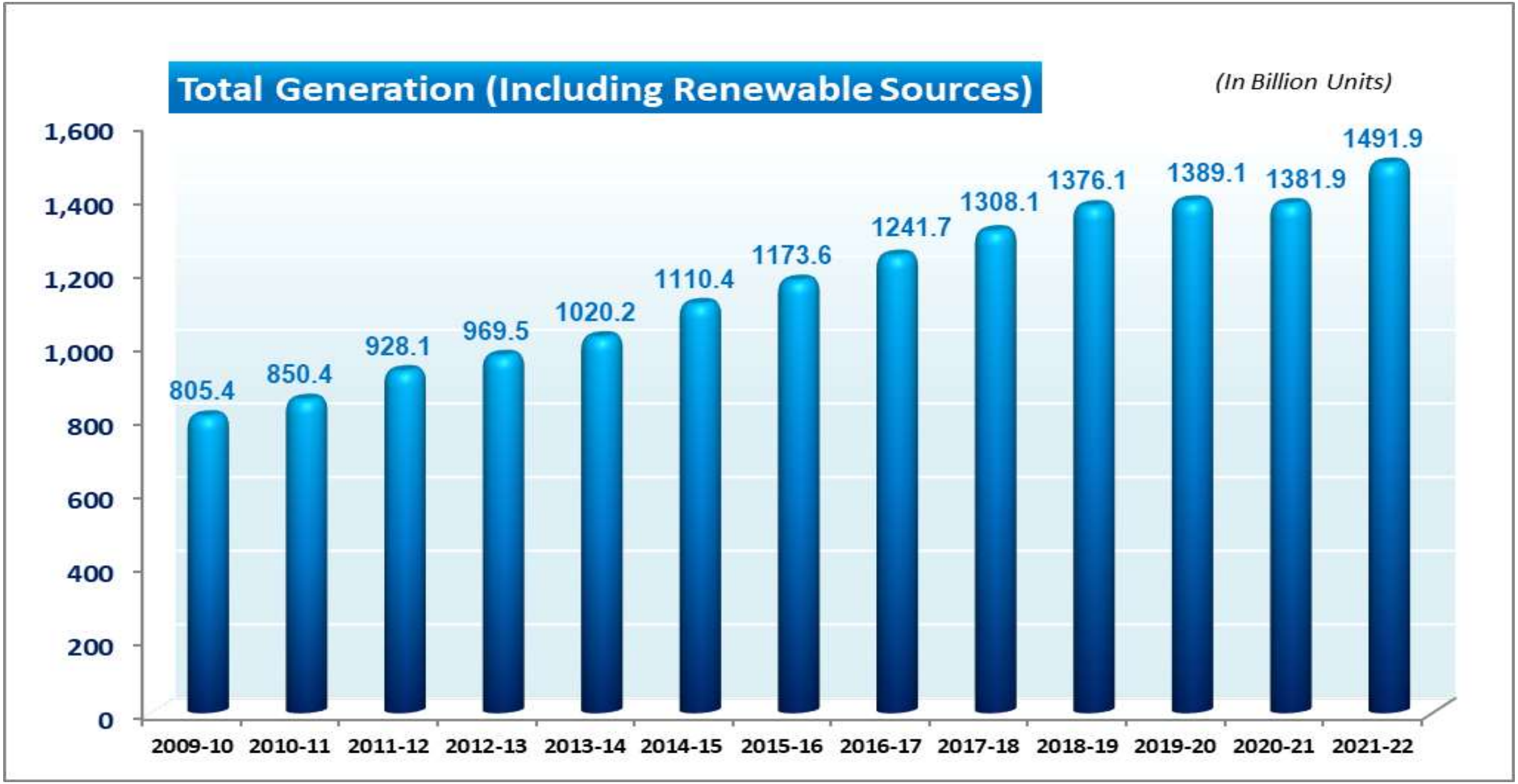
Plant Load Factor (PLF): The PLF in the country (Coal & Lignite based) from 2009-10 to 2022-23 is as under:

Year	PLF	Sector-wise PLF (%)		
	%	Central	State	Private
2009-10	77.5	85.5	70.9	83.9
2010-11	75.1	85.1	66.7	80.7
2011-12	73.3	82.1	68.0	69.5
2012-13	69.9	79.2	65.6	64.1
2013-14	65.60	76.10	59.10	62.10
2014-15	64.46	73.96	59.83	60.58
2015-16	62.29	72.52	55.41	60.49
2016-17	59.88	71.98	54.35	55.73
2017-18	60.67	72.35	56.83	55.32
2018-19	61.07	72.64	57.81	55.24
2019-20	55.99	64.21	50.24	54.64
2020-21	53.37	61.78	44.68	54.27
2021-22	58.87	69.71	54.50	53.62
2022-23*	69.50	79.74	67.56	62.34

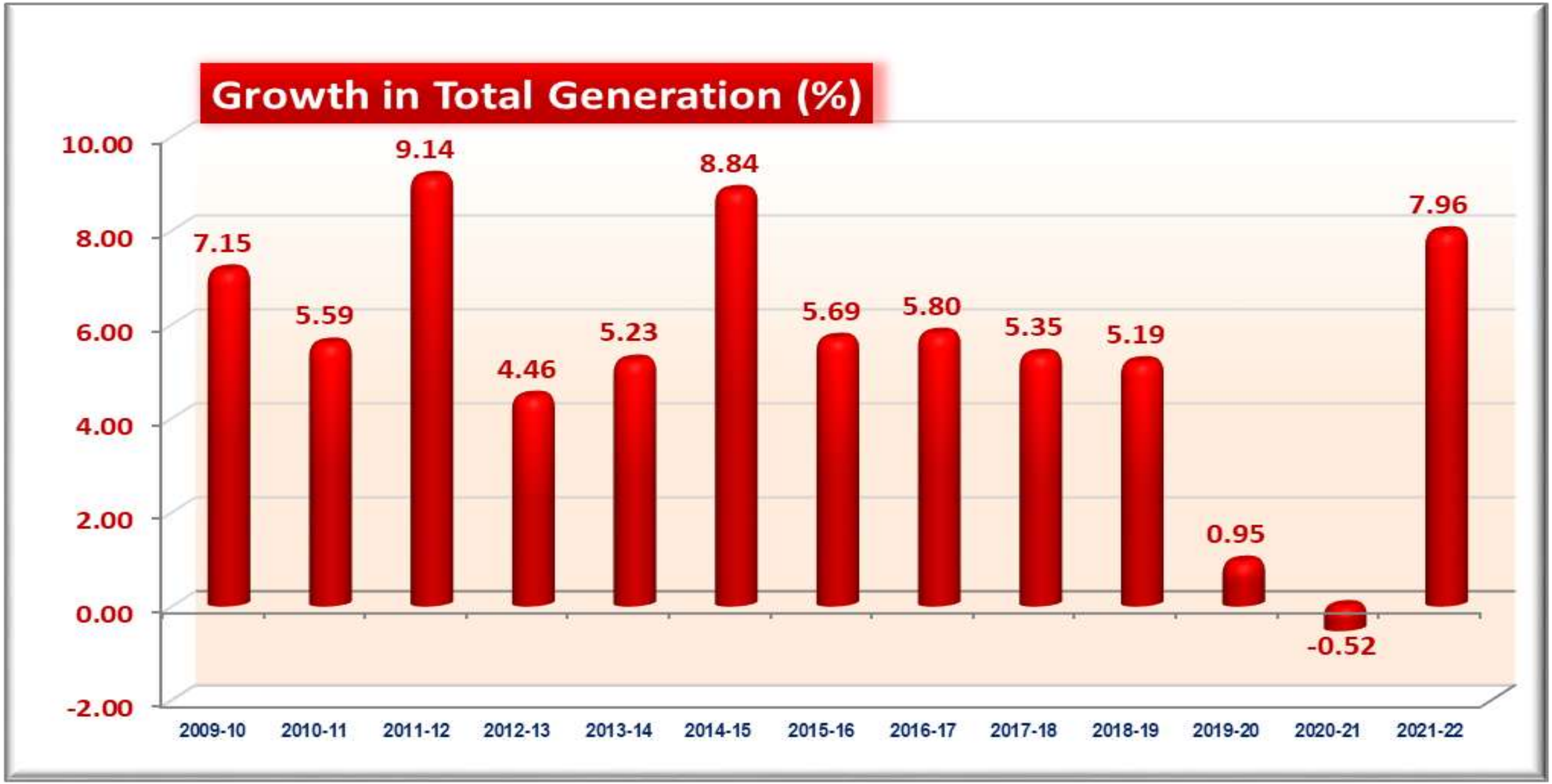
Power Supply Position The power supply position in the country during 2009-10 to 2022-23

Year	Energy				Peak			
	Requirement	Availability	Surplus(+)/Deficts(-)		Peak Demand	Peak Met	Surplus(+) / Deficts(-)	
	(MU)	(MU)	(MU)	(%)	(MW)	(MW)	(MW)	(%)
2009-10	8,30,594	7,46,644	-83,950	-10.1	1,19,166	1,04,009	-15,157	-12.7
2010-11	8,61,591	7,88,355	-73,236	-8.5	1,22,287	1,10,256	-12,031	-9.8
2011-12	9,37,199	8,57,886	-79,313	-8.5	1,30,006	1,16,191	-13,815	-10.6
2012-13	9,95,557	9,08,652	-86,905	-8.7	1,35,453	1,23,294	-12,159	-9.0
2013-14	10,02,257	9,59,829	-42,428	-4.2	1,35,918	1,29,815	-6,103	-4.5
2014-15	10,68,923	10,30,785	-38,138	-3.6	1,48,166	1,41,160	-7,006	-4.7
2015-16	11,14,408	10,90,850	-23,558	-2.1	1,53,366	1,48,463	-4,903	-3.2
2016-17	11,42,929	11,35,334	-7,595	-0.7	1,59,542	1,56,934	-2,608	-1.6
2017-18	12,13,326	12,04,697	-8,629	-0.7	1,64,066	1,60,752	-3,314	-2.0
2018-19	12,74,595	12,67,526	-7,070	-0.6	1,77,022	1,75,528	-1,494	-0.8
2019-20	12,91,010	12,84,444	-6,566	-0.5	1,83,804	1,82,533	-1,271	-0.7
2020-21	12,75,534	12,70,663	-4,871	-0.4	1,90,198	1,89,395	-802	-0.4
2021-22	13,79,812	13,74,024	-5,787	-0.4	2,03,014	2,00,539	-2,475	-1.2
2022-23*	4,04,761	4,00,654	-4107	-1.0	2,15,888	2,07,231	-8,657	-4.0

Generation (Billion Units)



Generation Growth (%)



What is Sustainable Development?

Sustainable development is an approach that aims to meet the needs of the present generation while preserving the ability of future generations to meet their own needs. It recognizes the interdependence between economic growth, social well-being, and environmental protection. Sustainable development recognizes that economic, social, and environmental considerations are interconnected and must be balanced to achieve long-term well-being.

Importance of Sustainable Development

- Focuses on Sustainable Farming Methods
- Manages to Stabilize the Climate
- Provides for Vital human needs
- Sustainable Biodiversity
- Financial stability